ERC Starting Grant 2026
Part B2[1]
(<u>not</u> evaluated in Step 1)

Scientific objectives

Previous studies have shown that early sensory manipulations can alter multisensory properties in the superior colliculus[1,2], and that reinforcement[3] or short-term repeated pairing of stimuli from different modalities can reshape cortical responses[4]. Yet, a systematic investigation directly connecting alterations in environmental cross-modal statistics during postnatal development to their long-term perceptual and neural outcomes in the cortex is still missing. MUSIC will fill this gap by exploiting advances in controlled audiovisual rearing, quantitative rodent psychophysics, two-photon imaging at neural population and synaptic levels, and computational modeling to provide the first integrated account of how cross-modal inference is learned, implemented, and expressed in cortical circuits. The overarching goal of MUSIC is to test whether unsupervised audiovisual experience during early postnatal life establishes cross-modal priors (i.e. expectations about one sensory modality conditioned on another) that (1) impact adult perception, (2) reshape visual cortical representations and spontaneous activity, and (3) leave measurable structural traces in synaptic architecture. These goals map onto three Work Packages (WPs) described in the following section.

Work packages

WP1 - *Impact of unsupervised audiovisual exposure on perception and behavior*

Methods. This work package aims to test whether prolonged exposure to a sensory environment with an experimentally imposed co-occurrence of visual orientation and auditory frequency can bias perception in adulthood. To achieve this goal, we will combine the following experimental approaches:

i) Altered rearing: Juvenile mice (>P20) will be continuously housed in automated audiovisual rearing systems consisting of modified cages with transparent, anti-reflective Plexiglas walls surrounded by computer monitors and overhead speakers. Each unit will be enclosed within a ventilated cabinet and connected to a central computer controlling stimulus presentation and logging. Animals will experience ∼10 h/day of intermittent audiovisual stimulation, respecting sleep-wake cycles and requiring minimal manual intervention. To generate an arbitrary but learnable audiovisual correspondence between orientation and sound frequency, natural images (e.g., from the Van Hateren Natural Image Database[5]) will be converted into corresponding soundscapes based on their orientation content. Orientation energy at a reference spatial frequency (~0.04 cycles/degree, matching the peak sensitivity of mouse V1 neurons[6]) will be quantified using a bank of oriented filters (e.g., Gabor filters). Only images with spatially uniform orientation content will be retained to ensure a location-independent mapping. The energy at each dominant orientation will then be linearly mapped onto sound frequency within the mouse-audible range (8-16 kHz[7]), establishing consistent correspondences (e.g., horizontal-rich ↔ low pitch, vertical-rich ↔ high pitch). This procedure yields a simple and reproducible algorithm for generating frequency to orientation correspondences between natural scenes and sound; arbitrary enough to leave a distinguishable signature in perception and neural activity yet structured enough to be learnable (Figure 1A). Conceptually, it draws inspiration from sensory-substitution paradigms for the visually impaired[8], adding translational relevance.

ii) 2AFC psychophysics: After reaching adulthood (>P60), animals will be trained in a freely moving two-alternative forced-choice (2AFC) orientation discrimination task to test whether the audiovisual correspondences experienced during rearing can induce cross-modal priors that bias visual perception. The task will be performed in a custom-built behavioral arena equipped with a central stimulus presentation monitor, two lateral reward ports, speakers, and overhead high-speed cameras for position and pose tracking. Animals will be trained to self-initiate trials by positioning themselves at the centre of the arena and orienting their body toward the monitor to trigger stimulus presentation. After observing the stimulus, mice will be free to move to one of the two reward ports to report their choice. During training, water-restricted animals will learn both the initiation rule and the task contingency (e.g., horizontal grating ↔ left port, vertical grating ↔ right port), attempting to maximise their water reward intake. After initial shaping with only the two extreme orientations (horizontal and vertical gratings), expert mice will be introduced to a

[1] Instructions for completing Part B2 can be found in the '*Information for Applicants to the Starting and Consolidator Grant 2026 Calls'*.

continuum of ambiguous stimuli in a subset of trials, making discrimination between orientations more difficult (e.g., low-contrast gratings or orientations near 45°). In some trials, auditory stimuli of different frequencies will be presented simultaneously to assess their influence on visual judgements. We expect these sounds to bias orientation perception according to the learned orientation-frequency association, particularly in low-discriminability trials. The behavioral effect will be quantified by analysing shifts in psychometric curves[9] for visual choices in the presence versus absence of sound (Figure 1B), and by fitting generalised linear models (GLMs) incorporating both auditory and visual predictors.

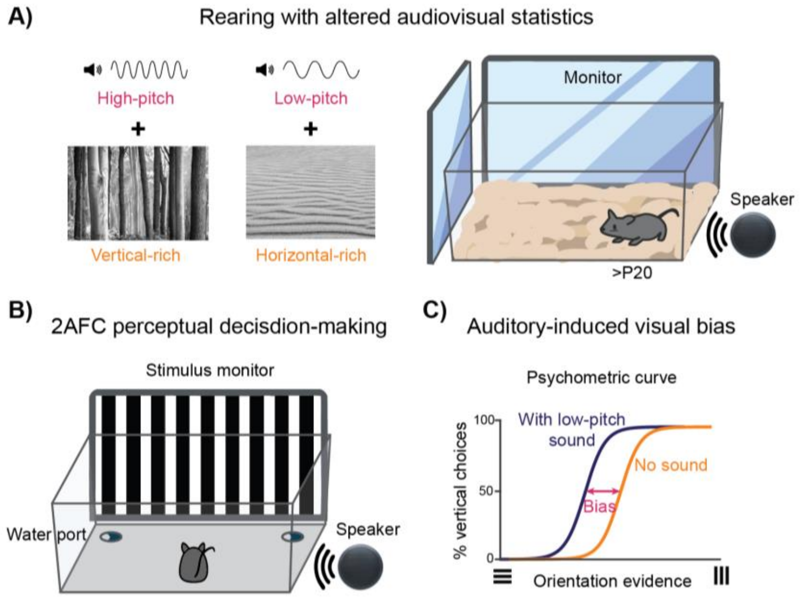


Figure 1 | WP1: Impact of unsupervised audiovisual exposure on perception and behavior
A) Rearing under altered audiovisual statistics. Post-weaning juvenile mice will be raised in custom-built arenas where naturalistic visual scenes are paired with algorithmically generated soundscapes. Visual orientation content is systematically mapped onto sound pitch. B) Perceptual testing in adulthood. After rearing, adult mice will perform a freely moving two-alternative forced-choice (2AFC) task to discriminate visual grating orientation while auditory stimuli are played back. C) Predicted perceptual effect. If cross-modal priors have formed, auditory cues consistent with the rearing contingencies will bias perceptual judgements toward the associated visual orientation, shifting psychometric curves.

iii) Fear conditioning: An operant-free second-order cross-modal fear conditioning[10] assay will test whether sounds alone can elicit freezing in darkness after visual conditioning. During the first-order conditioning phase, mice will be placed in an arena equipped with monitors displaying an oriented grating (CS1, e.g., horizontal) paired with a mild foot shock (US). On the following day, the same visual stimulus will be presented without shock to confirm conditioning efficacy by measuring freezing behavior. In the second-order test phase, we will assess whether the auditory stimulus (CS2) that was consistently associated with the conditioned visual stimulus during rearing (e.g., the low-frequency sound paired with the horizontal grating) can also evoke freezing in the absence of visual input. Mice will be placed in complete darkness, and the corresponding sound will be presented while freezing is recorded through automated video analysis. Generalisation of freezing to the paired auditory stimulus would provide an alternative readout of learned multisensory associations, confirming that rearing-induced audiovisual correspondences influence fear responses even when only one modality is available.

Animal cohorts and feasibility. A total of ~30 mice (mixed-sex) will be used to assess the perceptual effects of altered audiovisual rearing, and an additional 20 mice for the fear-conditioning assay. Each experiment will include an experimental cohort, exposed to consistent audiovisual pairings before behavioral testing as described above, and a control cohort reared with identical exposure to the same unimodal auditory and visual stimuli but with randomised cross-modal pairings. This will enable us to isolate the specific contribution of audiovisual structure. The PI has extensive experience in rodent altered-rearing paradigms[11], 2AFC psychophysics[12], and behavioral automation[13], ensuring efficient training and robust performance, particularly in freely moving conditions, which reduce stress and accelerate learning relative to head fixation. The shaping period required to an animal to reach expert performance and start psychophysical testing is expected to be 20-30 days. Multiple training setups will be built to enable parallel training of several animals potentially in several batches per day, maximising throughput, and reproducibility. Each rearing cabinet will host 4 large cages housing up to 5 mice per cage, maintaining social housing while ensuring uniform sensory exposure. This configuration will allow one full experimental and control cohort to be processed in parallel, enabling completion of the full behavioral dataset within ~3 rearing cycles.

Risks and contingency plan. A key question is whether prolonged unsupervised audiovisual exposure will produce measurable and specific behavioral effects. The heightened plasticity during the critical postnatal period[14], together with the prolonged duration of exposure, should work in our favour and maximise the likelihood of inducing robust multisensory learning. Furthermore, extensive literature demonstrates the effectiveness of postnatal sensory manipulations in producing strong changes in neural encoding, both in unimodal contexts[15] (including previous work by the PI[11]), and in multisensory systems such as the rodent superior colliculus and cortex[1,2,4]. While naturalistic image-to-sound mappings maximise ecological validity and introduce realistic stimulus variability, they may nevertheless prove too complex for robust and generalisable associative learning, particularly when tested with simpler visual stimuli such as gratings during the perceptual task. If this occurs, a simplified deterministic rearing protocol (e.g. horizontal grating ↔ low pure tone, vertical grating ↔ high pure tone) will be implemented, providing clear and easily generalisable correspondences (being very close to a paradigm has already been demonstrated effective[4]). If variability in 2AFC performance limits the sensitivity to detect perceptual biases, the second-order fear-conditioning assay will serve as an independent, operant-free validation of cross-modal learning[10]. These complementary behavioral paradigms will ensure that the impact of early audiovisual exposure can be reliably detected and quantified across multiple readouts.

WP2 - ***Impact of audiovisual exposure on evoked and spontaneous cortical activity***

Methods. This work package aims to test whether the behavioral biases established in WP1 are reflected in cortical population dynamics. Specifically, we will examine how audiovisual rearing modifies the neural representation of visual features in primary and associative visual cortices in the presence of rearing sounds (WP2.1), and whether these sounds can reinstate those representations in the absence of visual input (WP2.2). To achieve this goal, we will adopt the following experimental approaches:

i) Mini2P population imaging 2AFC behavior: We will perform freely moving two-photon calcium imaging (Figure 2A) using a lightweight (~3 g) Mini2P[16] microscope (Thorlabs, Figure 2B) while mice perform the 2AFC orientation-discrimination task established in WP1. Imaging will target layer 2/3 excitatory neurons in V1 and higher-order associative visual areas[17], such as the anterolateral area (AL), which is likely involved in audiovisual integration. As a proxy for neural activity, we will record fluorescence transients of the genetically encoded calcium indicator GCaMP6f[18], broadly expressed in these neurons by crossing Rasgrf2-A-2A-dCre and Ai148-D transgenic lines[19]. During surgery, mice will receive a ~4 mm cranial window over the target visual cortex. A custom head-fixation headbar and Mini2P baseplate will be implanted such that the glass window lies parallel to the cortical surface, using a stereotaxic manipulator for precise alignment. The headbar-baseplate assembly will be secured to the skull with surgical glue and dental cement, and the window will be protected post-operatively with a removable silicone cap. After recovery, mice will be head-fixed briefly to attach the Mini2P scope to the baseplate before each session. Upon completion of each session, the microscope will be detached and the protective cap replaced. The behavioral arena will be equipped with a cable-suspension system supporting the Mini2P fibre to allow the animals undisturbed movement. This setup enables high-yield, chronic population recordings during behavior, with up to several hundred neurons simultaneously imaged per 500 × 500 μm field of view (FOV) at a frame rate of 15 Hz (but up to 40 Hz with smaller FOV)[16]. Behavioral events, overhead video recordings, and imaging data will be synchronised across acquisition systems using National Instruments DAQ boards controlled by custom Matlab code. Mini2P image acquisition will be handled by ScanImage software, ensuring precise temporal alignment between behavioral and neural recordings. Using the neural population activity recorded while animals perform the task in the testing condition (i.e. with ambiguous stimuli and sound presented in a subset of trials, as described in WP1), we will train decoders to classify stimulus orientation under full-contrast visual conditions, then apply these decoders to low-contrast trials presented with or without rearing-consistent sounds. When such sounds are presented, population activity patterns are expected to shift toward the neural representation of the associated orientation, producing neurometric curve shifts that mirror the psychometric biases observed behaviorally in WP1 (Figure 2E). To further characterise these effects, we will apply representational similarity analysis (RSA) and low-dimensional embedding techniques (e.g. PCA, or t-SNE) to visualize and compare the geometry of activity evoked by visual and audiovisual stimulation across experimental and control cohorts[20].

ii) Mini2P population imaging during passive stimulation: To test whether playback of auditory cues in darkness can reactivate ensembles of visual neurons encoding the orientations associated with each sound during rearing, we will perform Mini2P recordings in the same animals, this time outside of the task (i.e. under passive conditions). Mice will freely explore a dark arena while rearing-consistent sounds are played at random intervals. This experiment directly tests the "*sampling theory*" interpretation of Bayesian inference in cortical dynamics[21,22], which posits that population activity at any given moment represents a sample from the posterior distribution (i.e. the combination of sensory evidence and prior expectation). In the absence of direct visual input, spontaneous population activity is therefore expected to represent the prior alone, here conditioned by the presence of an auditory cue that was consistently associated during rearing with a specific visual orientation. Under this framework, population activity in V1 during sound presentation should resemble that evoked by the corresponding visual stimulus. Decoders trained on high-contrast visual trials (from the behavioral sessions or from passive grating presentations in the same arena) will then be applied to these sound-only trials to determine whether auditory-evoked activity patterns in visual cortex resemble those elicited by visual stimulation (Figure 2F). A significant match would indicate that auditory input can drive cortical population states predictive of specific visual features, consistent with the formation of cross-modal priors during early development and with the "*sampling theory*" of cortical inference[21].

Animal cohorts and feasibility. For WP2, we will image 5-10 mice per group (audiovisual pairing vs. shuffled rearing control). Each mouse will contribute 10-20 imaging sessions across both behavioral (WP2.1) and passive (WP2.2) conditions, yielding several hundred sensory-responsive neurons per animal and ensuring sufficient statistical power for population analyses. The PI has extensive experience with chronic two-photon imaging, including pilot Mini2P recordings in freely moving mice, which have already demonstrated excellent optical stability, robust motion correction, high neuronal yield, and even preliminary evidence of multisensory neuronal selectivity in associative cortical areas (Figure 2A-D).

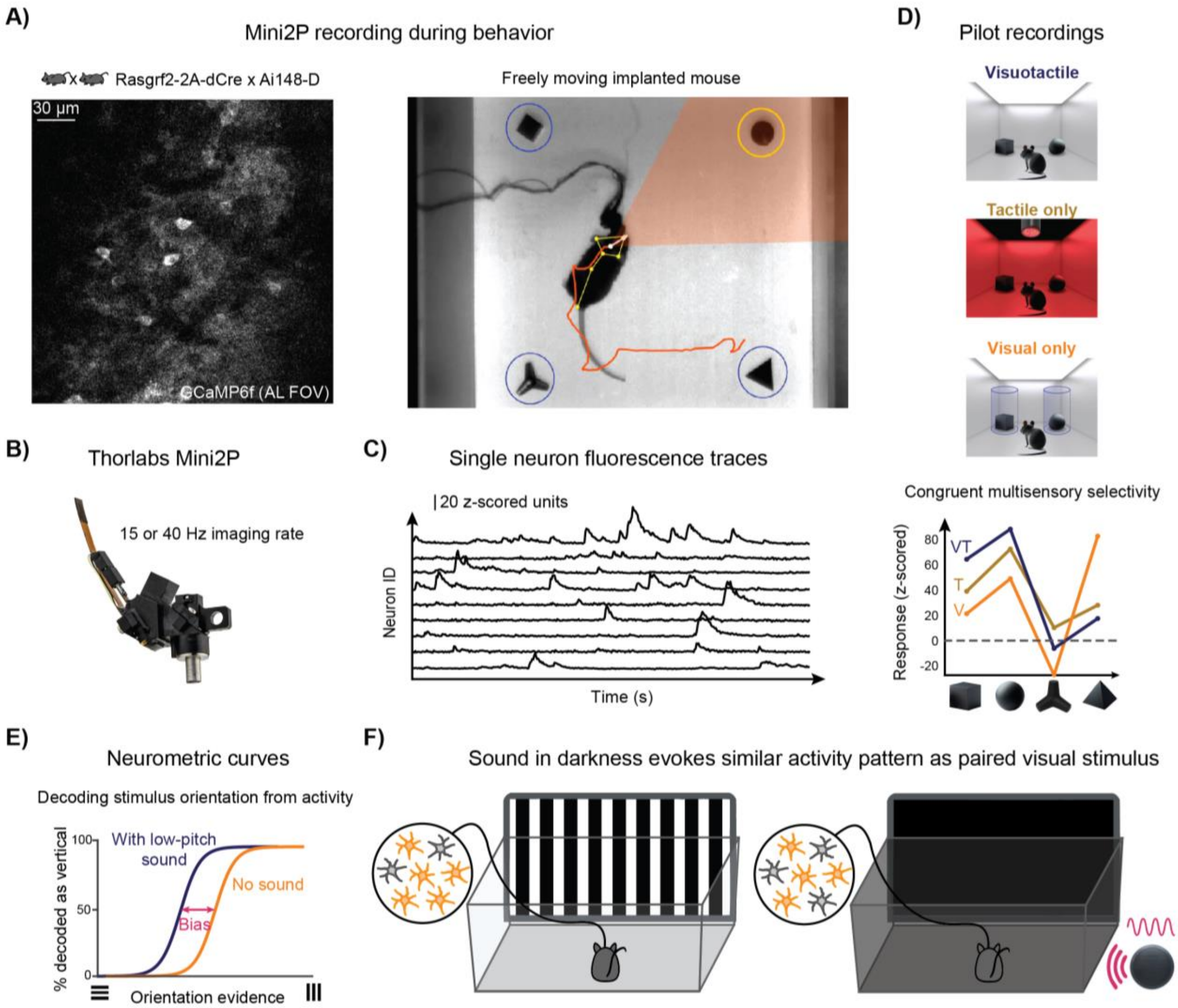


Figure 2 | WP2: Impact of audiovisual exposure on evoked and spontaneous cortical activity
A) Schematic of the freely moving Mini2P recording setup. Left: example of view (FOV). Right: behavioral tracking of a freely moving implanted mouse. B) Thorlabs Mini2P microscope used for imaging. C) Example calcium fluorescence traces from simultaneously recorded neurons during active exploration of objects. D) Multisensory object selectivity in associative cortex: matching responses to visuotactile, tactile only, and visual only stimulation. E) Schematic neurometric curves illustrating decoding of stimulus orientation from population activity (WP2.1). Testing the sampling hypothesis (WP2.2): in darkness, playback of rearing-consistent sounds is expected to evoke population activity patterns resembling those elicited by paired visual stimuli, consistent with sampling from learned priors.

Risks and contingency plan. A potential confound arises from movement-related activity contaminating sensory responses[23]. This issue will be addressed, as successfully done by the PI in previous studies, through high-speed behavioral video tracking and kinematic analysis to isolate genuine sensory encoding from movement-driven signals[13]. If freely moving passive-stimulation sessions required for testing the sampling theory hypothesis[21] (WP2.2) prove difficult due to variability in retinotopic positioning of stimuli, recordings could be performed under head fixation. This will exploit the headpost implanted with the Mini2P baseplate, ensuring ultra-precise control of stimulus retinotopic alignment. To obtain a higher temporal resolution view of population dynamics, chronic Neuropixels[24] recordings may complement Mini2P imaging. This approach would leverage the PI's extensive experience with in vivo extracellular electrophysiology[11,25,26] and will provide converging evidence across methodologies

WP3 - ***Synaptic and circuit substrates of audiovisual learning***
Methods. This work package aims to identify the synaptic correlates of audiovisual statistical learning, testing whether prolonged exposure to a specific audiovisual statistics leaves a measurable imprint on cortical microcircuits. Specifically, we will assess whether synaptic inputs representing associated auditory and visual features (i.e. those paired during rearing and therefore "congruent" for the animals) exhibit selective spatial clustering along dendritic branches of visual neurons. To achieve this goal, we will adopt the following approach:

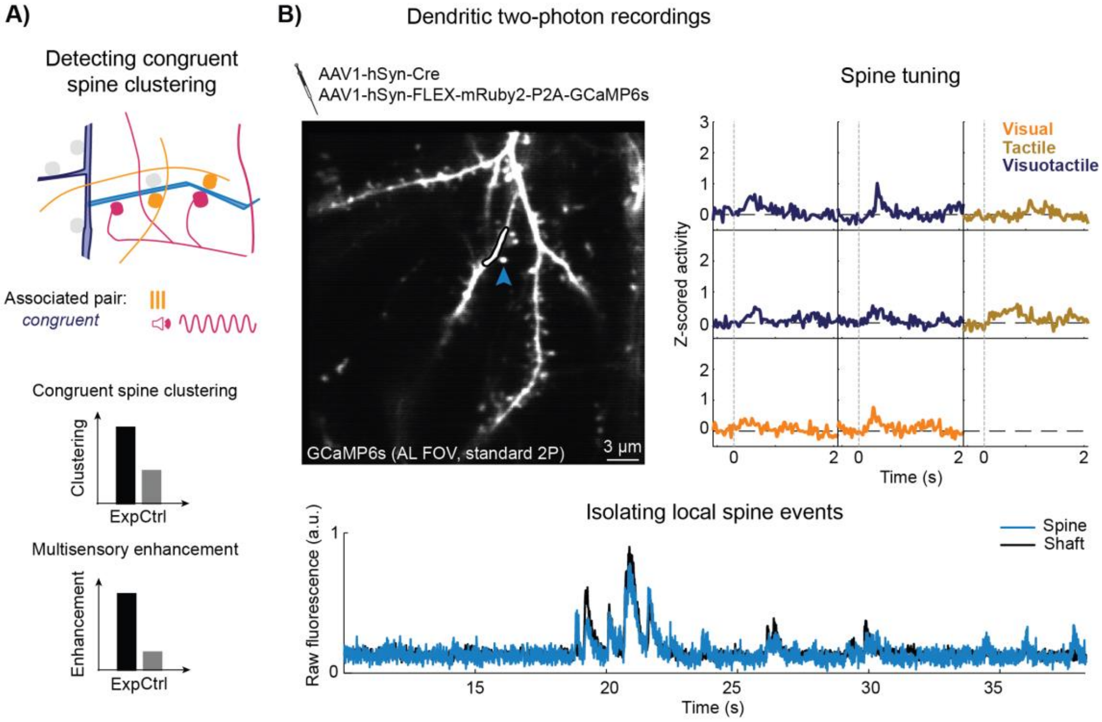


Figure 3 | WP3: Synaptic and circuit substrates of audiovisual learning
A) Conceptual schema illustrating the hypothesis that audiovisual rearing induces spatial clustering of congruent synaptic inputs . B) Example field of view from dendritic two-photon imaging in associative cortex from pilot experiments. Sparse GCaMP6s expression reveals dendritic spines suitable for functional analysis. Right: local spine responses showing tuning to visual (V), tactile (T), or visuotactile (VT) stimuli. Bottom: Example traces from a dendritic branch illustrating the isolation of local spine activity from shaft fluorescence.

i) Two-photon dendritic spine imaging: We will perform two-photon dendritic spine imaging in sparsely labelled visual neurons under head-fixed conditions during passive playback of auditory and visual stimuli using the Mini2P microscope. To obtain both functional readout and accurate motion correction, we will co-express GCaMP6s and a red fluorescent structural marker (e.g. mRuby) in a sparse subset of excitatory cortical neurons. Sparse expression is crucial to minimise neuropil contamination and allow reliable resolution of individual spines. Sparse single-cell labelling will be achieved by co-injecting a low-titre AAV vector expressing Cre recombinase together with a higher-titre AAV expressing a Cre-dependent GCaMP6s-mRuby[27]. Viral injections will be performed during window implantation. After full infection and stable expression (4-6 weeks post-surgery), head-fixed imaging sessions will be carried out in which visual and auditory stimuli are presented while local calcium transients from individual spines are recorded. These data will be used to extract each spine's tuning for orientation and sound frequency. To isolate locally driven synaptic signals from backpropagating somatic activity, we will regress out the fluorescence signal of the parent dendritic shaft from each spine's trace, retaining only local events[27]. By analysing the spatial distribution of tuning along dendritic segments, we will assess whether spines tuned to congruent auditory and visual features exhibit greater spatial clustering than those tuned to uncorrelated features. We expect spines tuned to congruent audiovisual features to cluster more tightly in reared animals than in controls, providing a structural correlate of associative learning. Such clustering would support cooperative dendritic nonlinearities[28] that selectively amplify congruent inputs, potentially giving rise to the phenomena observed in WP1-2 (Figure 3A). The degree of clustering enrichment will be quantified relative to randomized surrogate distributions within animals and across mouse cohorts.

Animal cohorts and feasibility. For WP3, we will image 10 mice per group (audiovisual pairing vs. shuffled rearing control). Each mouse will contribute multiple imaging sessions, yielding tens of dendritic segments and hundreds of dendritic spines per animal. This sample size will provide robust statistical power for assessing spatial clustering effects. The PI has prior experience with in vivo dendritic spine imaging using standard two-photon systems, including pilot head-fixed recordings that demonstrated clear optical access to fine dendritic structures and enabled the reconstruction of spine tuning during multisensory stimulation (Figure 3B).

Risks and contingency plan. Brain motion artefacts could compromise spatial precision and hinder reliable estimation of single-spine tuning properties. To mitigate this, co-expression of a structural marker will allow frame-by-frame motion correction via structural image registration, substantially improving spatial stability. This approach has already been successfully implemented by the PI in preliminary head-fixed recordings using a standard two-photon microscope (Figure 3B). A second challenge lies in isolating genuine synaptic signals from backpropagating action potentials. This will be addressed by regressing out fluorescence from the parent dendritic shaft to retain only locally driven events, a method that has already enabled reconstruction of visuotactile tuning in individual spines in pilot data (Figure 3B). If this correction proves insufficient, an alternative approach will involve expressing a glutamate sensor (e.g., iGluSnFR4[29]) to directly monitor synaptic release events, thereby circumventing the issue at its root.

CP - Cross-package computational modeling

Since MUSIC aims to quantitatively understand the neural mechanisms of multisensory learning, computational modeling will be tightly integrated with the main experimental work (WP1-3), both to generate a priori predictions and to link results to theory. We will build artificial neural network models of unsupervised multisensory learning that replicate in silico the same rearing and testing paradigms and analyses used experimentally, allowing direct comparison between model and data. The design "digital twins" can be varied to test specific hypotheses on learning rules or architectures[30]. A pilot implementation, developed by the PI in PyTorch and already available open source on GitHub, uses a contextual denoising autoencoder[31] (CDAE) implementing self-supervised learning, a form of unsupervised learning (Figure 4A).

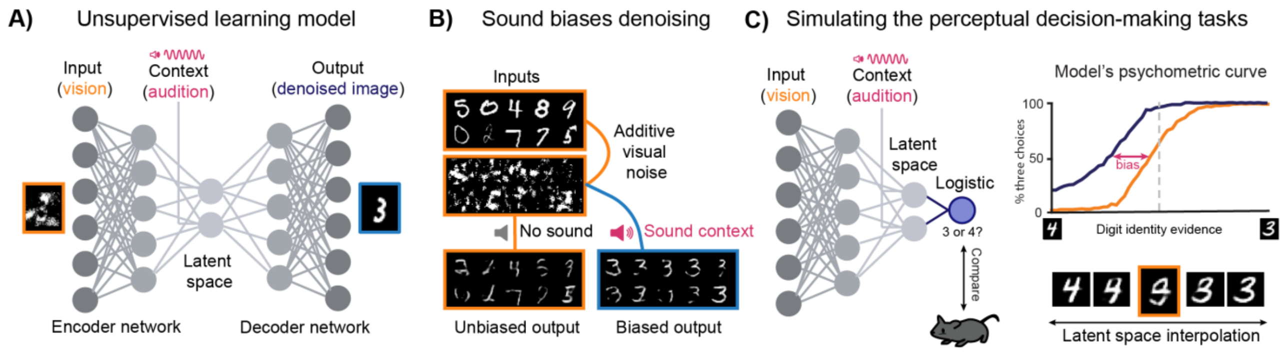
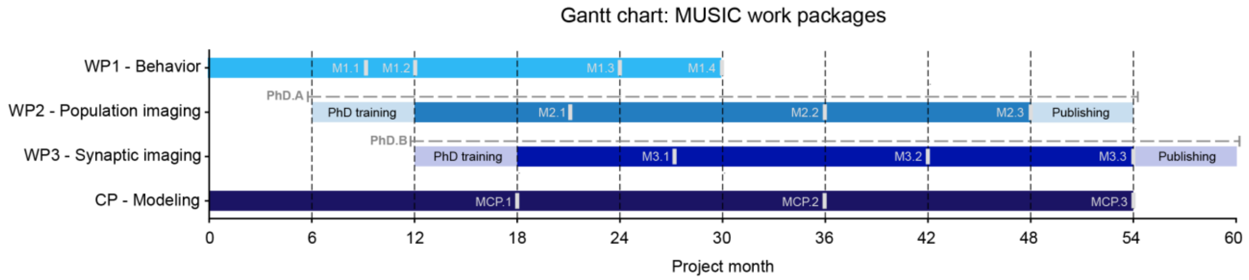


Figure 4 | CP: Cross-package computational modeling with artificial neural networks
A) Contextual denoising autoencoder (CDAE). The network receives a visual input and an auxiliary sound-context vector. During training, it learns to reconstruct clean images from noise-corrupted inputs, internalising input statistics. After visual pretraining, audiovisual fine-tuning is introduced, consistently pairing all instances of the digit "3" with the same "sound" to simulate the audiovisual rearing conditions that will be experienced by mice. B) Sound-induced reconstruction bias. When tested on noisy visual inputs, the presence of the associated sound biases the reconstruction toward the paired digit ("3"), revealing the emergence of a basic multisensory prior. C) Psychophysics of a contextual autoencoder. To mimic the 2AFC task, ambiguous visual inputs are generated by interpolating between latent representations of digits "3" and "4." A logistic classifier operating on these representations produces a psychometric-like curve showing a sound-induced shift in perceptual decisions.

This vision encoder-decoder network receives a visual input together with an auxiliary sound-context vector injected into the latent space. It is first trained as a visual denoising autoencoder[31], then fine-tuned with fixed audiovisual pairings mimicking the rearing contingencies. In the current setup, handwritten digits from MNIST[32] serve as visual inputs (to be replaced later with the rearing images that the mice will be exposed to), with all instances of a given digit (e.g. "3") paired with the same sound-context vector. The model successfully learns these cross-modal correspondences: auditory context biases visual reconstruction toward the associated digit (Figure 4B). When the decoder is replaced by a classifier and used for a binary digit classification ("3" vs. "4"), ambiguous stimuli (obtained by latent space interpolation) are perceptually "pulled" toward the rearing-consistent category, mirroring the predicted psychometric curve shift (Figure 4C). Although simplified, this approach captures the essential computational principle, unsupervised learning of cross-modal regularities, and provides a direct reference for interpreting the mouse data. Latent-space analyses of trained models will guide the interpretation of neural population activity, helping to understand whether cortical computations follow comparable statistical learning principles.

Work plan implementation

The project will be conducted by the PI and two PhD students (PhD.A and PhD.B) over a five-year period. The work plan is organised into three interconnected work packages (WPs), each with clearly defined milestones (M) and deliverables (D), as shown in the Gantt chart. Both PhD.A and PhD.B will contribute to WP1 activities, providing the foundation for the subsequent WPs. PhD.A will then concentrate primarily on population imaging (WP2), while PhD.B will focus on dendritic-level experiments (WP3). Both PhD students will begin training in parallel with WP1 to acquire the skills required for WP2 and WP3 execution.



WP1 (months 0-30): Milestones are M1.1, validation of the rearing and audiovisual exposure system (month 9); M1.2, validation of the 2AFC training procedure (month 12); M1.3, identification of cross-modal bias (month 24); and M1.4, assessment of conditioning generalization (month 30). Deliverables include fully operational audiovisual rearing and behavioral testing setups (D1.1), and the final analysis pipeline and behavioral dataset (D1.2).
WP2 (months 12-48): Milestones are M2.1, achievement of stable Mini2P recordings during behavior (month 21); M2.2, confirmation of neural correlates of audiovisual learning (month 36); and M2.3, testing of population sampling hypotheses (month 48). Deliverables comprise a complete two-photon population analysis pipeline and dataset (D2.1), including recordings during both active behavior and passive stimulation (D2.2). Gantt includes buffer time for PhD.A training and final publishing process (shaded).
WP3 (months 18-54): Milestones are M3.1, validation of the dendritic imaging pipeline (month 27); M3.2, validation of spine tuning assessment protocol (month 42); and M3.3, quantification of synaptic clustering (month 54). Deliverables include a validated dendritic analysis pipeline (D3.1) and the final dendritic spine imaging dataset (D3.2). Gantt includes buffer time for PhD.B training and final publishing process (shaded).
Cross-package modeling (months 0-54): Milestones are MCP.1, completion of the in-silico "digital twin" of behavioral experiments (month 18); MCP.2, completion of the in-silico "digital twin" of population imaging experiments (month 36); and MCP.3, final model integration and synthesis (month 54). Deliverables include the model architectures, trained weights, and full training and analysis codebase (DCP.1).

Justification for Additional Funding

In addition to the standard ERC StG budget (€1.5M), I request an additional €0.9M to cover the personnel, consumables, and equipment required to ensure full project feasibility. The proposed research combines extensive behavioural, imaging, and computational experiments, leading to high initial equipment and consumable costs, as well as substantial animal housing expenses. Two PhD students are essential to sustain the data collection and analysis workload across WPs, while the current EUR/CHF exchange rate further increases the effective costs. These additional funds are therefore crucial to secure the necessary staff, equipment and materials to guarantee timely, complete, and high-quality execution of the project.

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Appendix: All current grants and on-going / submitted grant applications of the PI (Funding ID)

Mandatory information (does not count towards page limits)

Current research grants (Please indicate "No funding" when applicable):

Project Title	*Funding source*	*Amount (Euros)*	*Period*	*Role of the PI*	*Relation to current ERC proposal*[2]
None	None	No funding	None	None	None

On-going / submitted grant applications (Please indicate "None" when applicable):

Project Title	*Funding source*	*Amount (Euros)*	*Period*	*Role of the PI*	*Relation to current ERC proposal*[2]
None	None	No funding	None	None	None

[2] Describe clearly any scientific overlap between your ERC application and the current research grant or on-going grant application.